CAN PROTOCOL

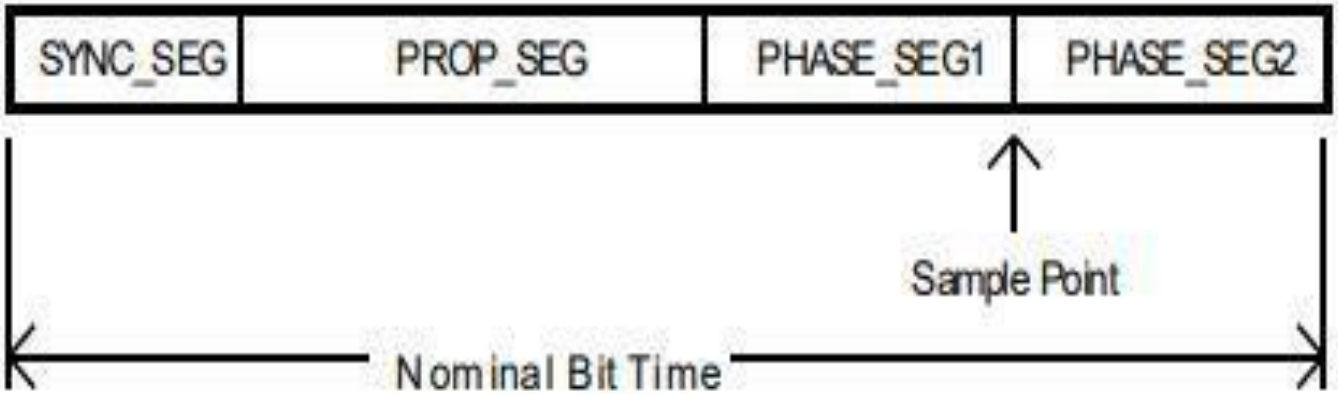
CAN BIT TIMING

CAN Bit Timing

- *Bit timing is the count of time Quanta(tq/Tq,a basic unit of bit time) required to carry a single bit(i.e tour of a bit on CAN bus from writing to reading) on CAN Bus.*
- *Reciprocal of Bit timing is known as Bit rate or Nominal bit rate(NBR).*
- *The Nominal Bit Rate (NBR) is defined in the CAN specification as the number of bits per second transmitted by an ideal transmitter with no resynchronization.*
- *Bit rate can be configured individually for each CAN node connected on CAN network.*
- *All nodes on a given CAN bus must have the same nominal bit rate.*
- *However, all devices are not required to have the same master oscillator clock frequency. For the different clock frequencies of the individual devices, the bit rate has to be adjusted by appropriately setting the baudrate prescaler and number of time quanta in each segment.*
- *The oscillator frequencies of Nodes(ECUs) have some instability because of some environmental factors which can cause synchronization error but every Node is configured itself to maintain its bit synchronization by adjusting bit time if the frequency deviation is within its tolerance limit. This adjustment is called Synchronization Jump Width(SJW) .*

- *Keeping the attention at bit synchronization a bit time is divided into 4 segments and each segment has some specific programmable number of time quanta(tq) –*

1. Synchronization Segment (**T**sync_seg)
2. Propagation Segment (**T**prop_seg)
3. Phase Segment 1(**T**phase_seg1)
4. Phase Segment 2(**T**phase_seg2)



Bit Time Components

- **Nominal Bit Time = Tsync_seg+TSEG1+TSEG2**

Where,

TSEG1 =Tprop_seg+Tphase_seg1

TSEG2 =Tphase_seg2

Tsync_seg=1 Tq

TSEG1 ,TSEG2 = N*tq(i.e integral multiple of tq)

- SAMPLE POINT

  The sample point is the point in the bit time at which the logic level is read and interpreted. The sample point is located at the end of PS1.
- A CAN system's propagation delay is calculated as being a signal's round-trip time on the physical bus (tbus), the output driver delay (tdrv) and the input comparator delay (tcmp).
- Assuming all nodes in the system have similar component delays, the propagation delay is explained mathematically:

$$tprop = 2 \cdot (tbus + tcmp + tdrv)$$

Segments	Values	Description
Tsync_seg	1 tq	Fixed
Tprop_seg	1..8] tq	Programmable.(Adjust according to physical delay time)
Tphase_seg1	[1..8] tq	Programmable.(Lengthened temporarily for synchronization)
Tphase_seg2	[2..8] tq	Programmable.(Shortened temporarily for synchronization)

- **Time Quanta(Tq)**

  Time quanta(tq) is the basic unit of the bit time and its formula is –

  tq=BRP/fsys

  BRP=Baud Rate Prescaler

  fsys =MCU system Clock(typically =Fosc)

- **Synchronization Segment**
- It is the first part of the bit time of fix length 1tq which Indicates the start of bit transmission. This segment is used to synchronize the transmission between nodes.
- If Bit transition occurs after **T**sync_seg then it is known as Positive phase error and this error will compensate by lengthening the **T**phase_seg1.
- If Bit transition occurs before **T**sync_seg then it is known as Negative phase error and this error will compensate by shortening the **T**phase_seg2.

Bit Rate Calculation Examples

Q. Calculate Required Time Quanta to achieve 500kbps baud rate if the system clock is 40mhz and CAN clock divider is 4.

- CAN clock(fcan)=40/4=10Mhz
- time Quanta(tq)=1/fcan=1/10Mhz=100ns
- bit rate =500kbps
- bit time=1/500kbps=1/500kHz=2us=2000ns
- no of tq=2us/100ns=2000ns/100ns=20tq.

Q. Configure TSEG1 and TSEG2 to set sampling at 80% of a bit time.

- **CAN Bit time(NBT)=Tsync+Prop+PS1+PS2,**
- **Here :TSEG1=Prop+PS1,TSEG=PS2**
- **NBT=Tsync+TSEG1+TSEG2**
- **Since Sample Point is 80% of NBT,Hence**
- (Tsync_seg+TSEG1)/(Tsync_seg+TSEG1+TSEG2)=80%
- as we calculated above: bit time = Tsync_seg+TSEG1+TSEG2=20tq
- so , (1+TSEG1)/(20)=80%
- TSEG1=16-1=15
- so TSEG2=NBT-Tsyn-TSEG1=20-1-15=4

so final configuration values are :

- Tsync_seg=1(fixed)
- TSEG1=15
- TSEG2=4

Referring MCP2515 Data Sheet

EQUATION 5-2:

$$TQ = 2 \cdot BRP \cdot T_{OSC} = \frac{2 \cdot BRP}{F_{OSC}}$$

Where: BRP equals the configuration as shown in Register 5-1.

5.3 Programming Time Segments

Some requirements for programming of the time segments:

- PropSeg + PS1 >= PS2
- PropSeg + PS1 >= TDELAY
- PS2 > SJW

For example, assuming that a 125 kHz CAN baud rate with FOSC = 20 MHz is desired:

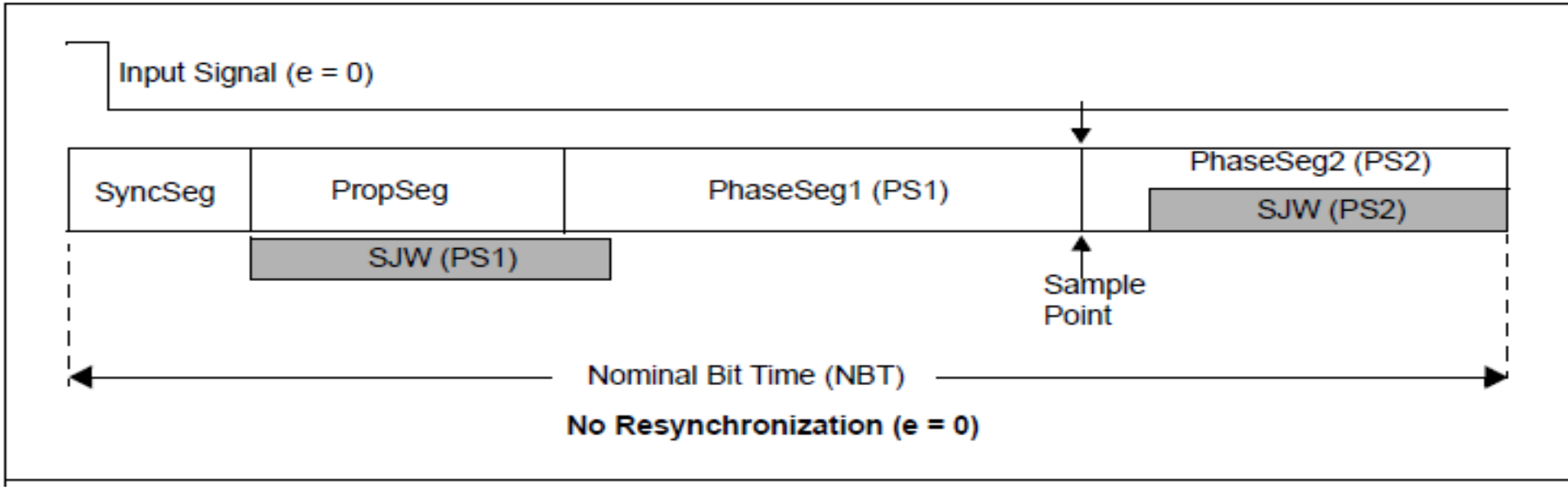
TOSC = 50 ns, choose BRP<5:0> = 04h, then TQ = 500 ns. To obtain 125 kHz, the bit time must be 16 TQ.

Typically, the sampling of the bit should take place at about 60-70% of the bit time, depending on the system parameters. Also, typically, the TDELAY is 1-2 TQ.

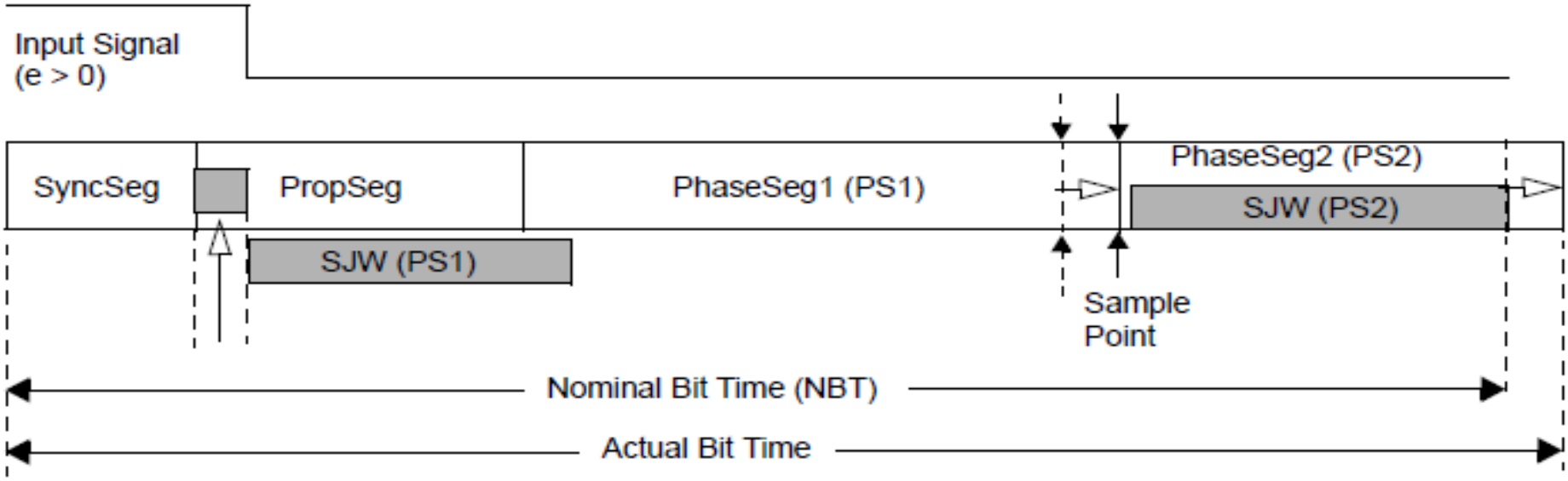
SyncSeg = 1 TQ and PropSeg = 2 TQ. So setting PS1 = 7 TQ would place the sample at 10 TQ after the transition. This would leave 6 TQ for PS2.

Since PS2 is 6, according to the rules, SJW could be a maximum of 4 TQ. However, a large SJW is typically only necessary when the clock generation of the different nodes is inaccurate or unstable, such as using ceramic resonators. So a SJW of 1 is usually enough.

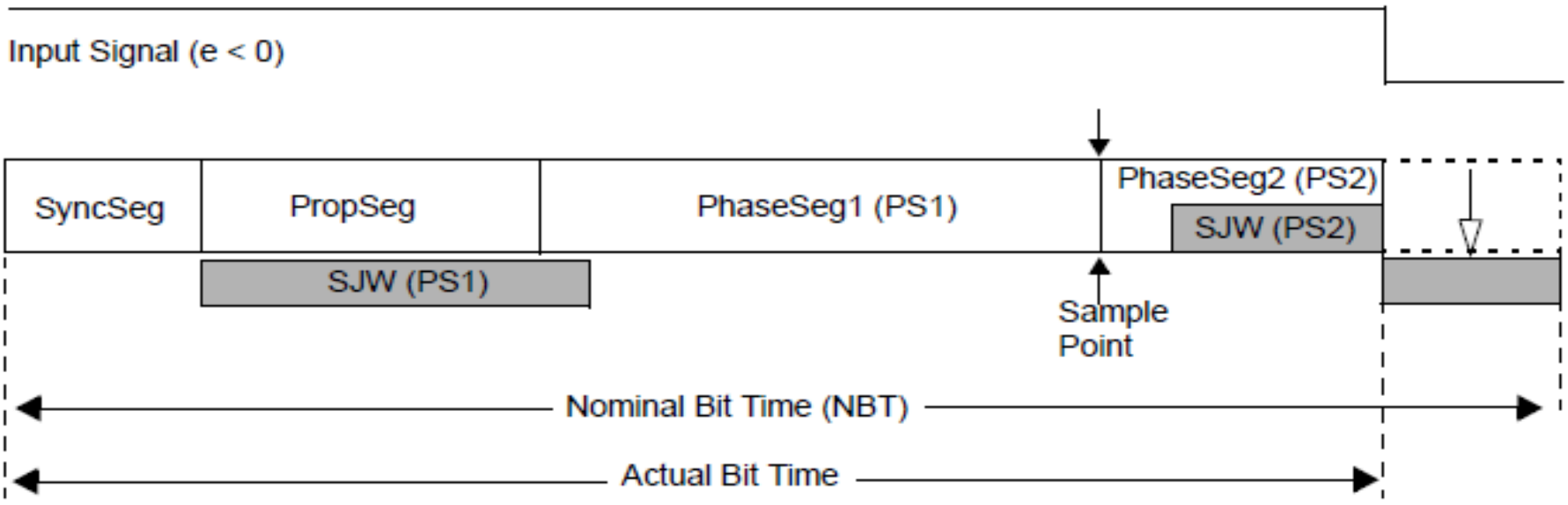
FIGURE 5-3: SYNCHRONIZING THE BIT TIME



No Resynchronization (e = 0)



Resynchronization to a Slower Transmitter (e > 0)



Resynchronization to a Faster Transmitter (e < 0)